
Exercises: Advanced Lesson 8

Your Name: Mikey Ferguson

Your Math Class: MATH 7013 – Advanced Mathematical Modeling

1: Part 1: Explain why it is usually better to use pattern matching for functional arguments.

Part 2: Give an example (other than the above) that fails when pattern matching is not used.

```
Clear[g, x, y];
```

```
g[x] = Sin[x];
```

```
{g[x], g[y], g[0], g[ $\frac{\pi}{2}$ ]}
```

```
Clear[g, x, y];
```

```
g[x_] = Sin[x];
```

```
{g[x], g[y], g[0], g[ $\frac{\pi}{2}$ ]}
```

Part 1: $f[x] = x^2$ attaches the definition to the single symbol x and to nothing else, so it is a stored value rather than a function. $f[x_] = x^2$ attaches it to the pattern any one argument, which is what a function of one variable means. Part 2: with $g[x] = \text{Sin}[x]$, $g[x]$ gives $\text{Sin}[x]$, but $g[y]$ and $g[0]$ come back unevaluated because they are different expressions.

2: Part 1: Explain why it is usually better to use delayed assignment for functional definitions.

Part 2: Give an example (other than the above) that fails when delayed assignment is not used.

```
Clear[h1, h2, x];
```

```
h1[x_] = Length[x];
```

```
h2[x_] := Length[x];
```

```
{h1[{1, 2, 3}], h2[{1, 2, 3}]}
```

Part 1: $=$ evaluates the right-hand side once, while the argument is still an empty symbol, and freezes that answer. $:=$ stores the right-hand side and re-evaluates it on every call. Part 2: $h1[x_] = \text{Length}[x]$ stores the length of the bare symbol x , which is 0, so $h1[\{1,2,3\}]$ gives 0. $h2$, defined with $:=$, gives the correct answer 3.

3: Construct a function $f[x]$ which:

Squares an integer argument,

Cubes a rational argument, and

Returns the same value for all other arguments.

Evaluate: $f[\frac{3}{5}]$, $f[3]$, $f[\frac{4}{2}]$, $f[3.1]$

Generate the following table of functional values:

```
values = Table[ { x, f[x] }, { x, -5, 5, 1/2 }]
```

Use ListPlot to visualize the functional values.

```
Clear[f, x, values];
```

```
f[x_Integer] := x2;
```

```
f[x_Rational] := x3;
```

```
f[x_] := x
```

```
??f
```

```
{f[ $\frac{3}{5}$ ], f[3], f[ $\frac{4}{2}$ ], f[3.1`]}
```

```
values = Table[{x, f[x]}, {x, -5, 5,  $\frac{1}{2}$ }]
```

```
ListPlot[values]
```

4: What is the difference between the Boolean tests: NumberQ and NumericQ

Evaluate both Boolean functions for: 3, 3/4, 3.1, 3+4*i*, $\sqrt{2}$, π , e

Which function would you choose for universal trigonometric evaluations and why?

```
Clear[vals];
```

```
vals = {3,  $\frac{3}{4}$ , 3.1`, 3 + 4 i,  $\sqrt{2}$ ,  $\pi$ ,  $e$ };
```

```
NumberQ /@ vals
```

```
NumericQ /@ vals
```

NumberQ asks whether something is literally stored as a number: it is True for 3, 3/4, 3.1 and 3 + 4 i, and False for Sqrt[2], Pi and E, which are stored as symbolic expressions. NumericQ asks the weaker question would this become a number under N, and is True for all seven. For trigonometry use NumericQ, because Pi and Sqrt[2] are exactly the arguments you want accepted.

5: What is the difference between the Boolean tests: VectorQ ListQ MatrixQ

Evaluate each of the Boolean functions for: 3, {1,2,3}, {{1,2},{3,4}}, {{{1},{2}},{3,4}}, {{{1,2},{3,4}}, {{5,6},{7,8}}}

```
Clear[objs];
```

```
objs = {3, {1, 2, 3}, {{1, 2}, {3, 4}}, {{{1}, {2}}, {{3, 4}}}, {{{1, 2}, {3, 4}}, {{5, 6}, {7, 8}}};
```

```
ListQ /@ objs
```

```
VectorQ /@ objs
```

```
MatrixQ /@ objs
```

ListQ is True for anything whose head is List, whatever its shape. VectorQ is True only for a flat list of elements, so only {1,2,3} passes. MatrixQ is True only for a rectangular list of lists, so only {{1,2},{3,4}} passes. The ragged example and the three-deep example are lists, but neither vectors nor matrices.

6: Define a function using Boolean tests which is:

= +Sin[x] when x is positive, and

= -Sin[x] when x is nonpositive.

Graph the function for $-\pi \leq x \leq \pi$

```
Clear[f, x];
f[x_] := Sin[x] /; Positive[x];
f[x_] := -Sin[x] /; !Positive[x]
Plot[f[x], {x, -π, π}]
```

7: Repeat exercise 3 using conditional evaluation i.e. "/;"

```
Clear[f, x];
f[x_] := x^2 /; IntegerQ[x];
f[x_] := x^3 /; Head[x] === Rational;
f[x_] := x
{f[3/5], f[3], f[4/2], f[3.1`]}
ListPlot[Table[{x, f[x]}, {x, -5, 5, 1/2}]]
```

8: Repeat exercise 6 using conditional evaluation i.e. "/;"

```
Clear[f, x];
f[x_] := Sin[x] /; x > 0;
f[x_] := -Sin[x] /; x ≤ 0
Plot[f[x], {x, -π, π}]
```

9: Repeat exercise 3 using the Which command.

```
Clear[f, x];
f[x_] := Which[IntegerQ[x], x^2, Head[x] === Rational, x^3, True, x]
{f[3/5], f[3], f[4/2], f[3.1`]}
ListPlot[Table[{x, f[x]}, {x, -5, 5, 1/2}]]
```

10: Repeat exercise 6 using the Which command.

```
Clear[f, x];
f[x_] := Which[x > 0, Sin[x], True, -Sin[x]]
Plot[f[x], {x, -π, π}]
```

11: Repeat exercise 3 using a conditional application of a rule.

```
Clear[x, rule3];
rule3 = {n_Integer → n^2, n_Rational → n^3, n_ → n};
```

$$\left\{-\frac{3}{5}, 3, \frac{4}{2}, 3.1\right\} /. \text{rule3}$$

$$\text{Replace}\left[\left\{-\frac{3}{5}, 3, \frac{4}{2}, 3.1\right\}, \text{rule3}, \{1\}\right]$$

$$\text{ListPlot}\left[\text{Table}\left[\{x, \text{Replace}[x, \text{rule3}]\}, \left\{x, -5, 5, \frac{1}{2}\right\}\right]\right]$$

12: Repeat exercise 6 using a conditional application of a rule.

```
Clear[x, rule6];
```

```
rule6 = {n_ :> Sin[n] /; n > 0, n_ :> -Sin[n]};
```

$$\{-1, -0.5, 0, 0.5, 1\} /. \text{rule6}$$

$$\text{Replace}\left[\{-1, -0.5, 0, 0.5, 1\}, \text{rule6}, \{1\}\right]$$

$$\text{Plot}[\text{Replace}[x, \text{rule6}], \{x, -\pi, \pi\}]$$